

Session 2

Presentation Techniques



Slides that help you, a voice that carries, a demo that survives.

Part	Topic	Time
01	Slides: diagrams and demos, not paragraphs	7 min
02	Body language, voice and nerves	8 min
03	Live demos — always have a backup plan	6 min
04	Practice: present to a small group for peer feedback	9 min

Seven words you will hear today.

Word	Plain meaning
Visual aid	Anything the audience looks at while you talk: slides, a diagram, the product.
Block diagram	Boxes joined by arrows that show how the parts of a system connect.
Body language	What your posture, face, eyes and hands say without words.
Voice modulation	Changing speed, volume and tone so you never sound flat.
Filler words	“Um”, “like”, “so, yeah” — sounds that fill a pause.
Backup plan	What you’ll do if the live demo fails.
Peer feedback	Comments from classmates, given to help you improve.

A slide is a road sign, not a page.

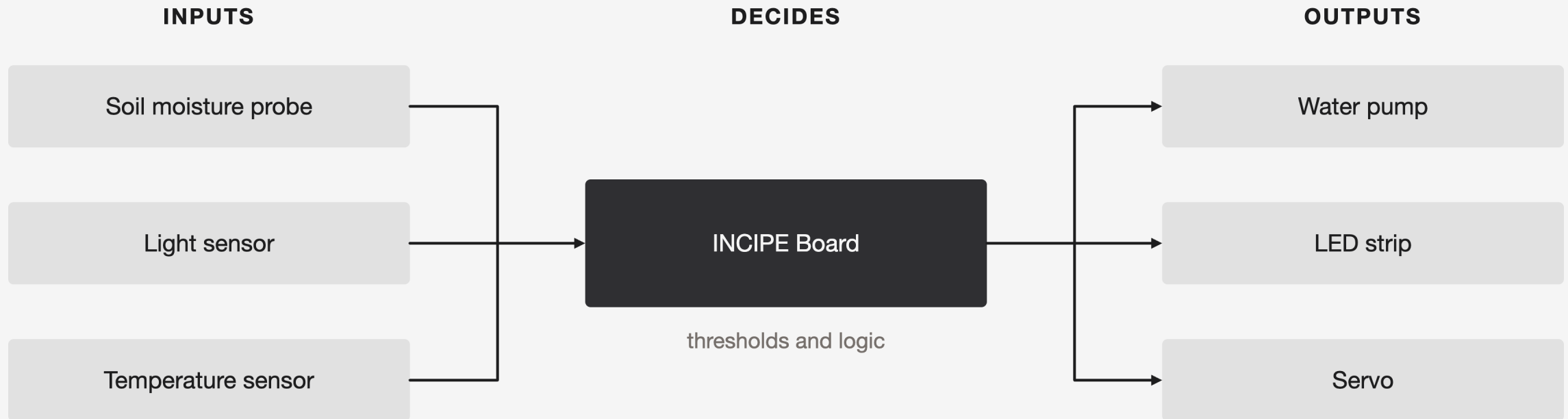
Drivers read a sign in a second: one symbol, a few words. Your audience is “driving” too — they’re listening to you.

If a slide is a paragraph, they read instead of listening, and lose both.

Five rules for slides that help.

Rule	Instead of...	Do this
One idea per slide	Five points on one slide	Five slides, one point each
Few words	A paragraph	A short phrase or a single number
Show, don't list	"Uses a probe, a board and a pump"	A block diagram: probe → board → pump
Big and readable	14-point text	Text the back row can read
Real pictures	Clip art of a plant	A photo of your build

Notice → decide → act, in one glance.



Plan a “Live demo” slide — and a copy right after it with a photo or video of the demo, in case you need it.

Four habits.

Do	Why
Stand still, weight on both feet	Swaying or pacing looks nervous and distracts.
Look at three people in different parts of the room, and rotate	Eye contact makes each listener feel spoken to.
Keep your hands visible; point at the slide or product	Hidden hands look unsure; open hands look confident.
Face the audience, not the screen	Read nothing from the slide — you already know it.

Speak like a song, not a single note.

Tool	How to use it
Pace	Slow down for the important parts — the problem and the impact.
Pause	Stop for two seconds after a key sentence. Silence makes people listen.
Volume	Speak to the back row — a little louder than feels natural.
Variety	Let your voice rise with a question and fall at the end of a statement.
Fillers	Replace “um” with a pause. Nobody notices a pause; everybody notices “um”.

Try it (2 min): read your first 30 seconds to a partner twice — flat and fast, then with pace, pauses and volume. Which would they remember?

Nerves are energy. Make them work for you.

Before	During
Rehearse out loud at least three times — once standing, once in front of someone.	Know your first 30 seconds by heart; the nerves usually settle after.
Breathe slowly, with a longer breath out than in, for a minute.	Lost your place? Pause, look at your notes, carry on.
Check the room and equipment early.	Speak to a friendly face first.
Call it excitement, not fear.	Slow down — nerves make everyone speed up.

Never perform a trick for the first time on stage.

A magician practises a hundred times — and always has another trick ready.

Your demo needs both: rehearsal on the real setup, and a backup.

What can go wrong – and how to prevent it.

What can go wrong	Prevent it
A sensor reads –1 (“not detected” on the Wiki)	Plug everything in and check the Serial Monitor before you start.
The board isn’t connected, or the Wi-Fi is different	Test on the same setup, in the same room. Bring a USB cable.
The board runs an old version	Upload the final, tested sketch the day before. Don’t change it on the day.
The demo needs exact conditions	Bring dry soil and a cup of water; a cloth to cover the light sensor.
Something just breaks	Have a backup.

The demo checklist.

#	Before you present
1	Record a 30–60 second video of the demo working. That's your backup.
2	Put it on the slide right after “Live demo”.
3	Plan the main path only: the one thing that proves your idea — dry soil → PUMP ON.
4	Start from a known state: board powered, Serial Monitor open, soil dry.
5	Practise the demo out loud, including what you say while it runs.

If it fails live: stay calm, say what should happen (“When the soil is dry, the pump switches on”), then switch to the backup. Audiences remember how you handled it.

Present to a small group. Groups of 3–4.

Role	Job
Presenter	Gives the 5-minute pitch from Session 1, with slides and a demo (or its backup).
Timekeeper	Shows a sign at 4 minutes and stops the presenter at 5:30.
Listeners	Fill in the glow-and-grow form while listening.

Glow and grow: one thing that worked, one to improve.

Area	Glow	Grow
Story: was the user the hero?		
Slides: one idea each, readable?		
Voice: pace, pauses, volume?		
Body: eye contact, stance, hands?		
Demo: did it prove the idea? A backup?		

“It was good” helps nobody. Be specific. Then fix your two most common grows before Session 3.

Checkpoint 1

A slide has 80 words about how the Smart Garden works. How do you fix it?

READ THE CODE

[80-word paragraph]

ANSWER

Diagram

WHY

Replace the paragraph with a block diagram (probe → board → pump) and a short title — and say the explanation.

Before a demo, what do you check so a -1 reading doesn't surprise you?

READ THE CODE

```
Serial Monitor:  
soil = -1.00
```

ANSWER

Sensors

WHY

That every sensor is plugged in and gives a real value. On the Wiki, -1 means “not detected”.

Checkpoint 3

The demo fails in front of the class. What do you do first?

READ THE CODE

```
PUMP ON ... ?  
(nothing happens)
```

ANSWER

Stay calm

WHY

Say what should have happened, then switch to your backup video or photos.

Checkpoint 4

You speak too fast when nervous. Name one technique.

READ THE CODE

```
"So-um-basically-it-waters-  
the-plant-and-um..."
```

ANSWER

Pause

WHY

Mark pauses in your script after key sentences; breathe out slowly before you start.

Checkpoint 5

Feedback says only “It was good.” How do you make it useful?

READ THE CODE

Feedback: It was good.

ANSWER

Glow & grow

WHY

One specific thing that worked and one specific thing to improve.

Three things to take away.

1

Slides support you: one idea each, few words, diagrams instead of paragraphs, a demo slide with a backup after it.

2

Body and voice carry the story: steady stance, eye contact, pauses, variety — and rehearsal turns nerves into energy.

3

Rehearse the demo on the real setup, show the main path only, keep a backup video, and improve with glow-and-grow feedback.

Next — Session 3: Technical presentation skills. Explaining your sensors, code and system design to people who've never seen a circuit.